

Yin-Chi Chan, Ph.D.

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Summary

Data scientist / software engineer with a PhD and 10+ years of research experience in the simulation modelling of stochastic systems, spanning telecommunications and healthcare service delivery.

Former postdoctoral researcher and research software engineer, building simulation models, interactive dashboards, and data-driven tools to support operational decision-making in healthcare systems.

Works mainly in Python, with working experience in TypeScript, Rust, C++, and R – always open to picking up new languages and frameworks.

Currently based in the Toronto area, seeking industry roles in data science, simulation, or backend/data-focused software engineering.

Skills

Languages: Python, R, Rust, TypeScript, C / C++

Frameworks & Libraries: Dash, FastAPI, React

Simulation: SimPy, salabim, Arena, MATLAB / Simulink / Simscape

Tools: Docker, Kubernetes, Git, GitHub Actions, Pytest / Tox

Data: SQL, openpyxl, Pandas, NumPy, R tidyverse

Experience

University of Cambridge, Research Associate

Institute for Manufacturing (IfM)

Cambridge, UK

Dec 2021 – Mar 2026

4 years 4 months

- Supervisor: Prof. Ajith K. Parlikad
- Built discrete-event simulation (DES) models to analyze clinical lab workflows, enabling key performance indicator (KPI) estimation and scenario testing to support operational decision-making by NHS stakeholders, in collaboration with stakeholders at the Cambridge University Hospitals (CUH) NHS Trust.
- Designed DES models to assist operational decision-making at CUH, by predicting hospital bed demand caused by respiratory disease epidemics, including statistical distribution fitting for patient stay durations. Created a Python-based web interface (using Plotly and Dash) allowing non-technical CUH staff to run simulations and explore scenarios independently.

University of the Witwatersrand, Co-lecturer

School of Electrical and Information Engineering

Johannesburg, South Africa

(remote)

July 2021 – Sept 2021

3 months

- Co-lectured the capstone course "Design Project" on engineering design for Electrical and Information Engineering students, invited via prior collaboration with Prof. Anton van Wyk on epidemiological modelling for COVID-19.
- Students choosing the selected topic designed a simple epidemiological model for COVID-19, analyzed possible tipping points leading to an out-of-control local epidemic, and designed an early-warning system to avoid hitting those tipping points.

City University of Hong Kong, Research Associate

Developed stochastic and simulation models and approximation methods for healthcare and telecommunications systems, focusing on capacity planning and optimisation of cellular systems and intensive care networks.

Hong Kong

Mar 2020 – Oct 2021

1 year 8 months

- Supervisor: Dr. Eric W. M. Wong

City University of Hong Kong, Research Fellow

Hong Kong
Oct 2018 – Mar 2020
1 year 6 months

City University of Hong Kong, Postdoctoral Fellow

Hong Kong
Feb 2017 – Oct 2018
1 year 9 months

Education

PhD	City University of Hong Kong , Electrical Engineering - Dissertation: Surrogate-Based Approximation of Blocking Probability in Non-Hierarchical Overflow Loss Systems - Advisor: Dr. Eric W. M. Wong; Co-supervisor: Prof. Moshe Zukerman	Hong Kong Feb 2017
MSc	City University of Hong Kong , Multimedia Information Technology Distinction	Hong Kong July 2011
BMath	University of Waterloo , Mathematics Distinction	Waterloo, ON, Canada June 2010

Projects

Respiratory viruses modelling web app

Interactive Python dashboard for modelling respiratory-virus-induced hospital bed demand, developed in collaboration with Cambridge University Hospitals NHS Foundation Trust.

- Built using salabim for discrete-event simulation and Dash/Plotly for the web interface.
- GitHub: github.com/yinchi/virus-sim-dashboard

Data integration for space-aware digital twins of hospital operations

Python code for the Automation in Construction paper of the same name, predicting the effect of building layout and infrastructure status on turnaround time in a clinical laboratory.

- Built using ifcopenshell to process building geometry and salabim for discrete-event simulation; tracks entity decomposition/aggregation, batching/unbatching, and transport across the entire laboratory process.
- Documentation: yinchi.github.io/histopath-bim-des

Ring puzzle with solver and web UI

Clone of a classic ring puzzle with a solver and both terminal-based and web-based interfaces.

- Uses bidirectional depth-first search to precompute an endgame table, with parallelism and Python free-threading for efficient computation.
- Uses GitHub Actions to automatically build and deploy the web interface to GitHub Pages and run unit tests.
- Web UI: yinchi.github.io/ring_puzzle

CodeCrafters projects

Personal implementations of popular software, tested against CodeCrafters' automatic submission system. Site profile: app.codecrafters.io/users/yinchi

- HTTP server (Rust): HTTP/1.1 request handling with concurrency and persistent connections.
- Interpreter (Rust): tokenizer, parser, and interpreter for the Lox programming language.
- LLM client (Python): basic command-line LLM client using the OpenAI Python API.

Movielens Project

Movie recommendation system for partial completion of the HarvardX course "Data Science, Capstone".

- Implements Funk's matrix factorization algorithm, using RcppArmadillo to integrate R with the Armadillo C++ linear algebra library.
- Project report: yinchi.github.io/harvardx-movielens

Continuing Education

Machine Learning Specialization

July 2026

Professional Certificate (DeepLearning.AI, in collaboration with Stanford University) covering supervised learning, advanced learning algorithms, and unsupervised/recommender/reinforcement learning.

Mathematics for Machine Learning and Data Science

May 2026

Professional Certificate (DeepLearning.AI) covering the mathematical foundations behind machine learning and data science techniques.

Engineering Design and Simulation

Apr 2026

Professional Certificate (MathWorks) using Simulink and Simscape to model and simulate physical systems such as a quadcopter and a quarter-car suspension model.

Introduction to Kubernetes

Aug 2024

Certificate (LFS158x, The Linux Foundation) on Kubernetes fundamentals and container orchestration.

Data Science

Oct 2022

Professional Certificate (HarvardX) covering the full data science workflow, from data wrangling through statistical inference and machine learning, in R.

Epidemics — Origins, Spread, Control and Communication

Mar 2021

Professional Certificate (HKUx, University of Hong Kong) on the origins, spread, control, and public communication of infectious disease outbreaks.

Introduction to Quantum Computing

Feb 2020

Certificate (MIT xPRO) covering the foundations of quantum computing, including qubits, quantum gates, and quantum algorithms.

Publications

Future directions in space-aware hospital digital twins development and implementation

Sept 2025

N. Moretti, A. Mukherjee, **Y.-C. Chan**, M. Nakaoka, A. K. Parlikad
doi.org/10.1049/icp.2025.3672 (Low-Cost Digital Solutions for Industrial Automation 2025)

Data integration for space-aware Digital Twins of hospital operations

Aug 2025

N. Moretti, **Y.-C. Chan**, M. Nakaoka, A. Mukherjee, J. Merino, A. K. Parlikad
doi.org/10.1016/j.autcon.2025.106276 (Automation in Construction)

Design, Performance Evaluation and Optimization for Intensive Care Networks Based on Non-Hierarchical Overflow Loss Systems

Mar 2025

J. Wu, W. Tian, A. Wang, **Y.-C. Chan**, E. W. M. Wong, K. Chan, G. Joynt
doi.org/10.1109/jbhi.2025.3549142 (IEEE Journal of Biomedical and Health Informatics)

A Process Simulation Model for a Histopathology Laboratory Y.-C. Chan , A. Mukherjee, N. Moretti, M. Nakaoka, J. Merino, Z. Rosun, C. Carr, D. McFarlane, A. K. Parlikad doi.org/10.1109/wsc63780.2024.10838731 (2024 Winter Simulation Conference)	Dec 2024
Dynamics of a susceptible-infected-recovered model on complex networks with inter-regional migration R. Niu, Y.-C. Chan , E. W. M. Wong, M. A. Wyk, S. Liu doi.org/10.1103/physreve.110.024304 (Physical Review E)	Aug 2024
Integrating teletraffic theory with neural networks for quality-of-service evaluation in mobile networks Y.-C. Chan , J. Wu, E. W. Wong, C. S. Leung doi.org/10.1016/j.asoc.2023.111208 (Applied Soft Computing)	Feb 2024
A Simulation Model and Dashboard for Predicting Covid-19 Bed Requirements Y.-C. Chan , K. Dreesbeimdiek, A. K. Parlikad, T. Ridgman, N. J. Matheson, B. Warne, D. Franks doi.org/10.1109/wsc60868.2023.10407602 (2023 Winter Simulation Conference)	Dec 2023
Low-cost digitalisation opportunities in healthcare: a histopathology department case study N. Moretti, A. Mukherjee, Y.-C. Chan , G. Yilmaz, J. Merino, M. Sasidharan, Z. Rosun, C. Carr, D. McFarlane, A. K. Parlikad doi.org/10.1049/icp.2023.1735 (Low-Cost Digital Solutions for Industrial Automation 2023)	Sept 2023
Healthcare process and variable mapping for identifying digitalisation opportunities and building a hospital digital twin A. Mukherjee, N. Moretti, Y.-C. Chan , G. Yilmaz, M. Sasidharan, J. Merino, Z. Rosun, C. Carr, D. McFarlane, A. K. Parlikad doi.org/10.35490/ec3.2023.270 (Proceedings of the 2023 European Conference on Computing in Construction and the 40th International CIB W78 Conference)	July 2023
A Century-Long Challenge in Teletraffic Theory: Blocking Probability Evaluation for Overflow Loss Systems With Mutual Overflow E. W. M. Wong, Y.-C. Chan doi.org/10.1109/access.2023.3283803 (IEEE Access)	June 2023
Modeling the Impact of Vaccination on COVID-19 and Its Delta and Omicron Variants J. Wang, Y.-C. Chan , R. Niu, E. W. M. Wong, M. A. Wyk doi.org/10.3390/v14071482 (Viruses)	July 2022
A stochastic SEIHR model for COVID-19 data fluctuations R. Niu, Y.-C. Chan , E. W. M. Wong, M. A. Wyk, G. Chen doi.org/10.1007/s11071-021-06631-9 (Nonlinear Dynamics)	Oct 2021
Evaluating Non-Hierarchical Overflow Loss Systems Using Teletraffic Theory and Neural Networks Y.-C. Chan , E. W. M. Wong, C. S. Leung doi.org/10.1109/lcomm.2021.3052683 (IEEE Communications Letters)	May 2021
Power Consumption and GoS Tradeoff in Cellular Mobile Networks With Base Station Sleeping and Related Performance Studies J. Wu, E. W. M. Wong, Y.-C. Chan , M. Zukerman doi.org/10.1109/tgcn.2020.3000277 (IEEE Transactions on Green Communications and Networking)	Dec 2020

Modeling the COVID-19 Pandemic Using an SEIHR Model With Human Migration R. Niu, E. W. M. Wong, Y.-C. Chan , M. A. Van Wyk, G. Chen doi.org/10.1109/access.2020.3032584 (IEEE Access)	Oct 2020
Performance Evaluation of 5G mmWave Networks with Physical-Layer and Capacity-Limited Blocking J. Wu, M. Wang, Y.-C. Chan , E. W. M. Wong, T. Kim doi.org/10.1109/hpsr48589.2020.9098993 (2020 IEEE 21st International Conference on High Performance Switching and Routing)	May 2020
Overflow models for the admission of intensive care patients Y.-C. Chan , E. W. M. Wong, G. Joynt, P. Lai, M. Zukerman doi.org/10.1007/s10729-017-9412-8 (Health Care Management Science)	Dec 2018
Blocking Probability Evaluation for Non-Hierarchical Overflow Loss Systems Y.-C. Chan , E. W. M. Wong doi.org/10.1109/tcomm.2017.2784450 (IEEE Transactions on Communications)	May 2018
Energy Efficiency-QoS Tradeoff in Cellular Networks with Base-Station Sleeping J. Wu, E. W. M. Wong, Y.-C. Chan , M. Zukerman doi.org/10.1109/glocom.2017.8254042 (2017 IEEE Global Communications Conference)	Dec 2017
Surrogate models for performance evaluation of multi-skill multi-layer overflow loss systems Y.-C. Chan , J. Guo, E. W. Wong, M. Zukerman doi.org/10.1016/j.peva.2016.06.007 (Performance Evaluation)	Oct 2016
Performance analysis for overflow loss systems of processor-sharing queues Y.-C. Chan , J. Guo, E. W. M. Wong, M. Zukerman doi.org/10.1109/infocom.2015.7218518 (2015 IEEE Conference on Computer Communications)	Apr 2015

Patents

Base station sleep mode based on power consumption and/or quality of service US Patent 10,568,047 B2. Assignee: City University of Hong Kong. - Inventors: W. M. E. Wong, J. T. Wu, Y. C. Chan, M. Zukerman patents.google.com/patent/US10568047B2	Feb 2020
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